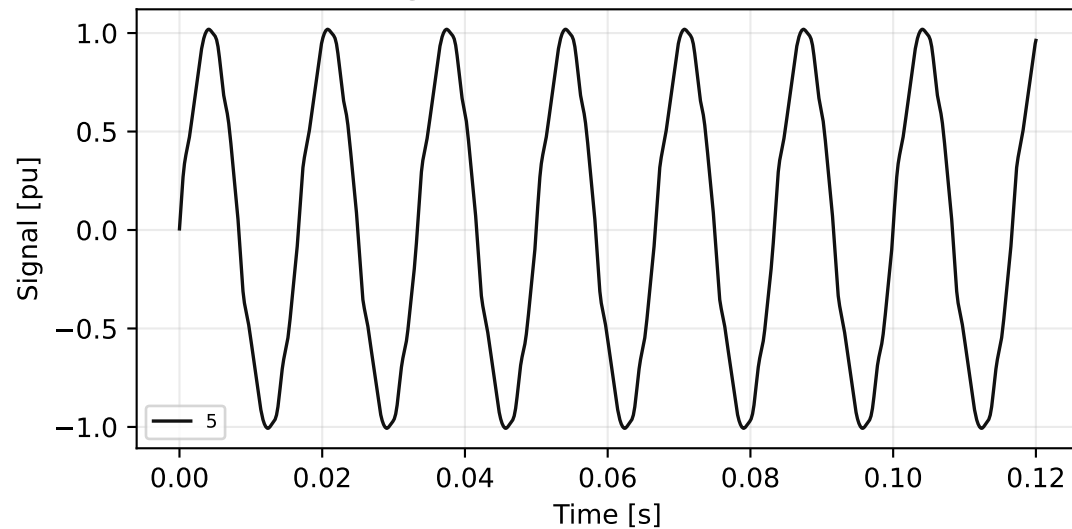


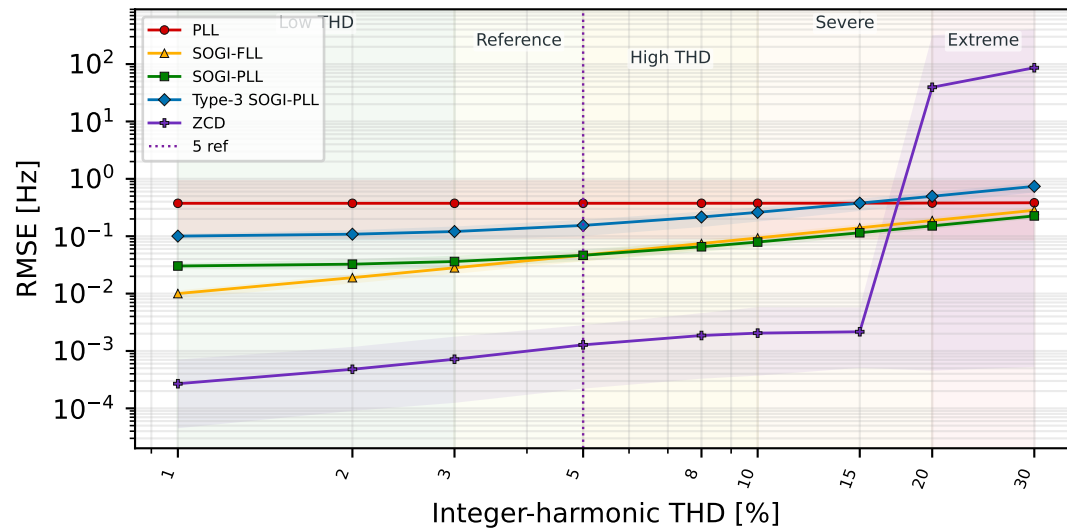
Integer Harmonics: RMSE [Hz] by estimator family (fixed policy)

Integer-harmonic isolation sweep: the true frequency is fixed at nominal and integer harmonic coefficients are scaled to a target THD. Interharmonics, subharmonics, impulses and frequency events are disabled so degradation can be attributed to harmonic distortion rather than mixed IBR artifacts.

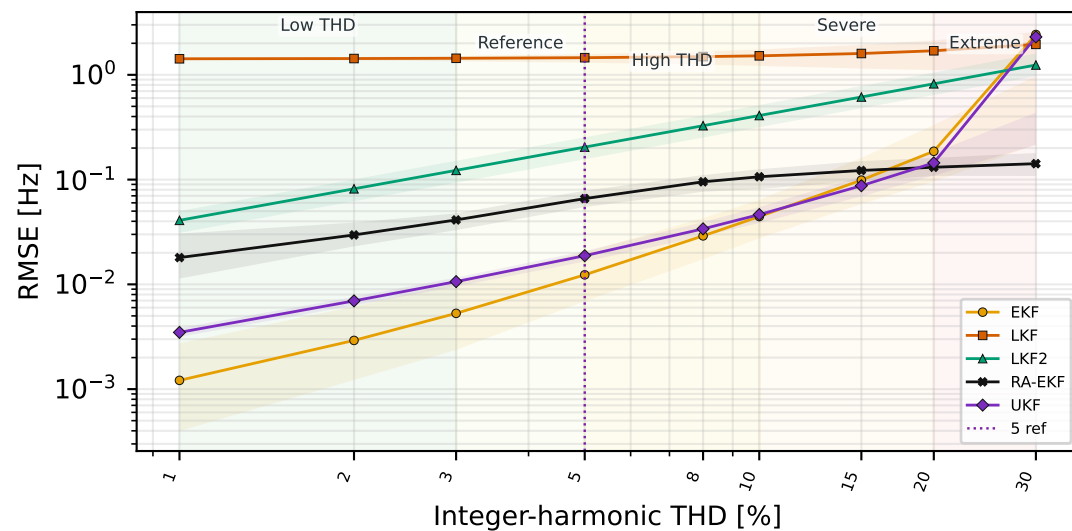
Reference Integer Harmonics



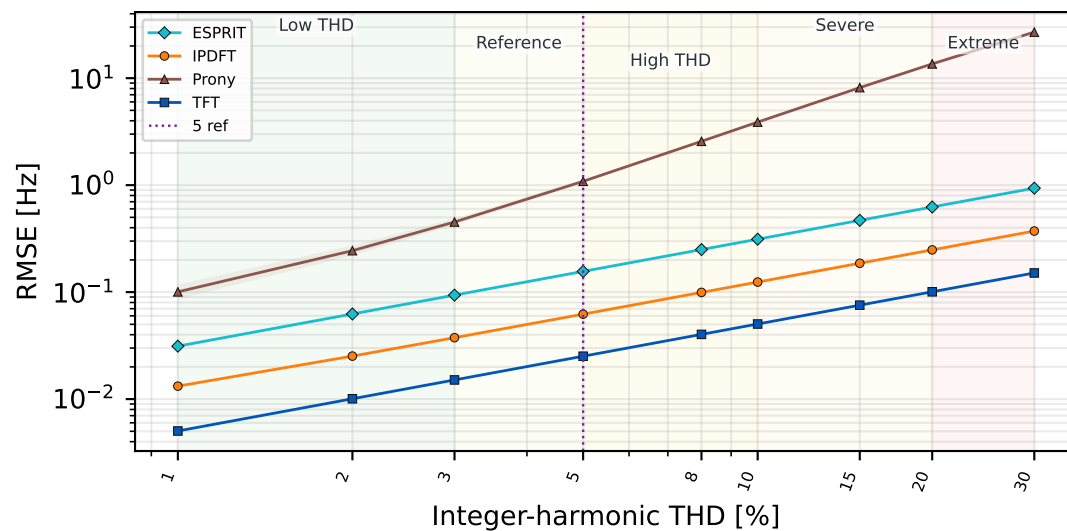
Loop-based



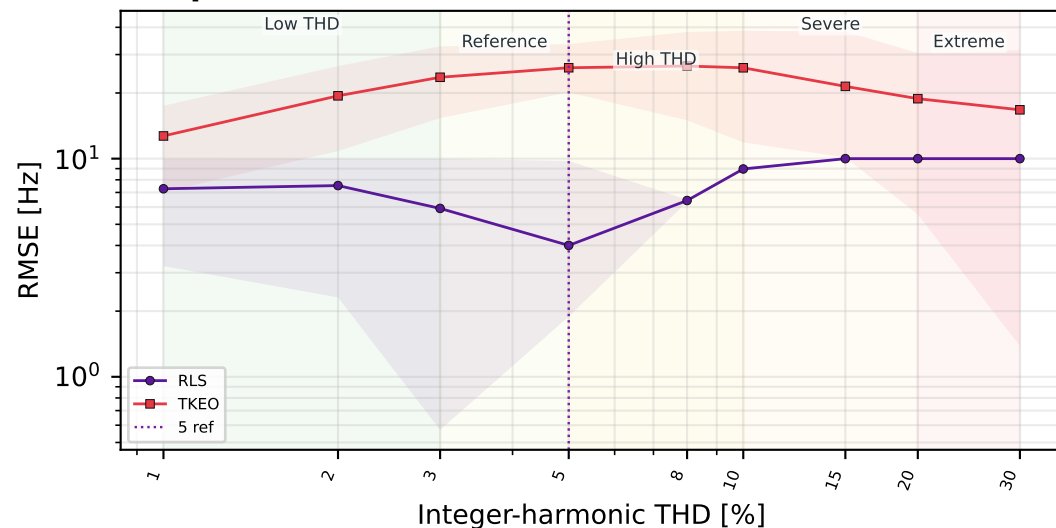
Model-based



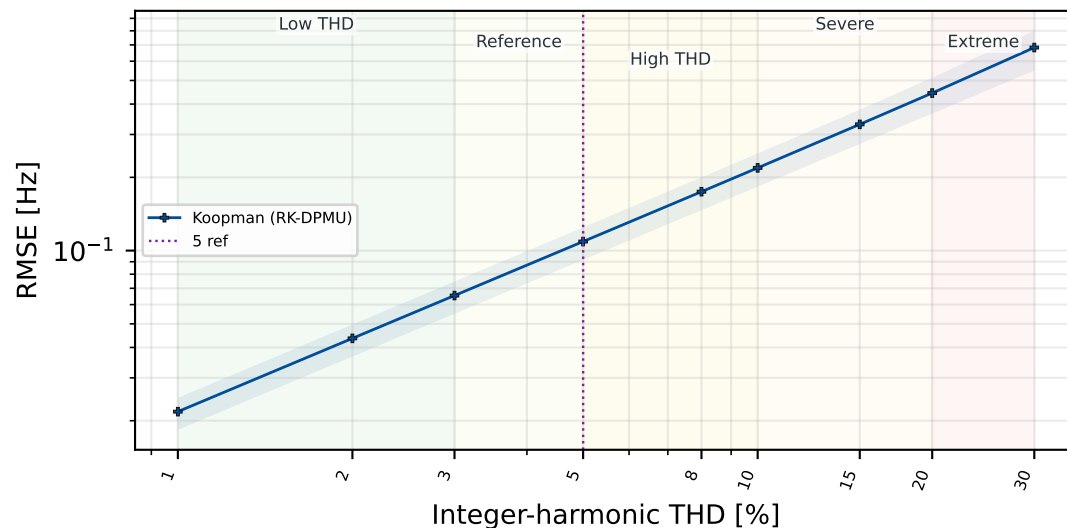
Window-based



Adaptive



Data-driven



Solid/dashed lines mark perturbation sign when applicable. Bands show available run intervals; thresholds and regions are interpretation guides.